

Summary of Test Outcomes and Generators

The full text dump of the pytest log is available in the zip file this document was in (it's about 35k lines). The file is called **test_results_fixed_errors.txt**.

Test pass/fail statistics:

Total tests failed: 915

Total tests passed: 1347

Total tests ran: 2262 (6 were skipped, 10 had warnings)

This is not a bad distribution of passing tests, as the programs in the dataset *were* buggy.

The AI agent generated some tests that had errors. About 10 test programs had incorrect import statements for directly running the dataset programs, which just required me to change the lines slightly and set up the dataset programs as actual python modules. It wasn't very hard to fix. The rest of the programs that didn't have these incorrect imports just copied the function under test line-for-line within the test python file, which I then verified were actually correct. I would have preferred all of the tests to use the original dataset program, but the agent decided to not do that so for the sake of the experiment I left it be.

Four test programs had other errors that I decided to fix so all of the tests would run. These were all related to using the `@example` annotation in Hypothesis, which defines a specific input to use for a testing function. These tests were intended to test for edge cases. However, Hypothesis errors when there are `@example` annotations and no `@given` annotations, so I just added in the missing `@gives`, with the strategy being any value the function accepts. This cleared the errors.

There were 21 test suites where most (all or all-but-one) test cases failed. I investigate these in more detail in the full-length report, but the short answer is that these were mostly caused by errors in the *formatting of the test cases*, *not* bugs in the original program.

Note: All of the generators I used were generated by the AI agent, I had nothing to do with them. I wanted to see how effectively the agent would define its own generators. This ended up causing some problems that I think are illuminating. See the full report.

I analyzed many more tests in detail in the full report.